Ex. No.: 5
Date:

System Calls Programming

Aim: To experiment system calls using fork(), execlp() and pid() functions.

Algorithm:

1. **Start**
   - Include the required header files (stdio.h and stdlib.h).
2. **Variable Declaration**
   - Declare an integer variable pid to hold the process ID.
3. **Create a Process**
   - Call the fork() function to create a new process. Store the return value in the pid variable:
     - If fork() returns:
       - -1: Forking failed (child process not created).
       - 0: Process is the child process.
       - Positive integer: Process is the parent process.
4. **Print Statement Executed Twice**
   - Print the statement:

     scss
     Copy code
     THIS LINE EXECUTED TWICE

     (This line is executed by both parent and child processes after fork()).

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Open format_program.c
~/

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
int main(){
        int pid;
        pid = fork();
        printf("This line executed twice\n");
        if(pid == -1){
                printf("Child process not created\n");
                        exit(0);
        }
        else if(pid == 0){
                printf("\n--- Child Process ---\n");
                printf("Child process Id (PID): %d\n", getpid());
                printf("Parent process ID (PPID): %d\n",getppid());
                printf("Executing 'ls' command using execlp()\n");
                execlp("ls", "ls", NULL);
        }
        else{
                printf("\n--- Parent Process ---\n");
                printf("Parent Process Id (PID): %d\n", getpid());
                printf("Parent's parent process Id (PPID): %d\n", getppid());
        }
        printf("It can be executed twice\n");
        return 0;
}
```

```
[deepika@localhost ~]$ vi format_program.c
[deepika@localhost ~]$ gcc format_program.c -o format_program
[deepika@localhost ~]$ ./format_program.c
bash: ./format_program.c: Permission denied
[deepika@localhost ~]$ ./format_program
This line executed twice

--- Parent Process ---
Parent Process Id (PID): 32679
Parent's parent process Id (PPID): 32536
It can be executed twice
This line executed twice

--- Child Process ---
Child process Id (PID): 32680
Parent process ID (PPID): 1941
Executing 'ls' command using execlp()
average.awk  Downloads       format_program.c  Public       shell_file
Deepika_045  employees.dat   mark.awk          reverse.sh   subject.dat
Desktop      fibo.sh         Music             salary.awk   Templates
Documents    format_program  Pictures          sample1.txt  Videos
[deepika@localhost ~]$
```